

For my final project I will use publicly available electricity system data from the Independent Electricity System Operator (IESO), an Ontario based energy grid dataset, which includes hourly renewable generation by fuel type, system demand, and related time series data. I will build machine learning models to forecast future wind and solar generation based on historical patterns in the renewable output and related features. These forecasts will be integrated into an optimization model that prescribes how a simulated energy storage asset should be operated at each hour to decide when to store energy or use it to meet demand. The optimization model will include constraints on storage capacity and charge or discharge limits and will be compared to baseline strategies to evaluate performance. I will measure the value of the forecast informed storage dispatch in terms of increased utilization of renewable generation and reduced system cost compared to simple rule based approaches. This project integrates prediction and prescriptive decision making using real energy data from Ontario's electricity grid and provides quantitative insights into how forecasts can improve operational decisions using storage and renewable generation. Sources of data include IESO hourly output by fuel type and historical demand series available from the IESO Power Data and Data Directory.

Link to Dataset:

<https://www.ieso.ca/Power-Data/Data-Directory#Variable%20Generation%20Forecast%20Summary%20Report>